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**VALUE-AT-RISK ESTIMATION USING GARCH
MODELS FOR THE CHINESE MAINLAND STOCK
MARKET**

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ABSTRACT

With the acceleration of economic globalization, the immature Chinese mainland stock market is gradually associated with the stock markets of other countries. This paper predict the return rate of Chinese mainland stock market using several models from GARCH family, test the predictability by calculating Value-at-Risk, also capture the dynamic correlation between other five countries or region and mainland China by DCC-GARCH model. The results indicate that E-ARMA-GARCH model fits the best due to the significant heteroscedasticity and leverage effect of Chinese mainland stock market. It has the strongest positive correlation with Hong-Kong while the weakest correlation with the United States.

Keywords: Volatility, Exponential-GARCH, VaR, Dynamic correlation

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1 Introduction

1.1 Background and Purpose

The stock market involves both speculators and investors, is a thermometer of a country or region's economic and financial activities. The stock market in mainland China, unlike ones such as the United States, Japan and the United Kingdom, is a capital flow market under the socialist market economy, which started relatively late and is still not perfect.

At the same time, with the increasing openness of global capital markets, the global stock markets are becoming more closely connected. The economy has maintained rapid and steady growth over the development of China's capital market in the past 30 years, the mainland stock market and the international market have been increasingly closely connected and influenced each other.

The first purpose is to predict the logarithmic return rate of Chinese mainland stock market and to test the predictability of the stock market return by analyzing the volatility. The analysis is based on the advantages of Autoregressive moving average (ARMA) model in forecasting stationary time series data and Generalized Autoregressive Conditionally Heteroskedastic (GARCH) model in the variability of financial data, constructing multiple models based on different information distributions. Investors expect the higher return and lower risk, thus Value-at-Risk (VaR) method is also employed in this paper, to measure the risk.

The Dynamic Conditional Correlation (DCC)-GARCH model has been widely used in the study of the correlation between international stock markets according to the existing research literature, it can also well capture the correlation between different stock markets. Therefore, based on the time-varying theory of DCC-GARCH model, this paper also studies the dynamic correlation between the Chinese mainland stock market and other five countries or regions.

The empirical results show that the daily logarithmic return rate of Shanghai composite index, which represents the stock market of mainland China, has significant heteroscedasticity, risk premium and leverage effect. Hence, the results suggest that among the adopted GARCH family, the E-ARMA-GARCH model follows skewed-student-t distribution performs

the best. Among the five countries and regions, Hong-Kong stock market has the most stable and strongest positive correlation with Chinese mainland market, Japan has the most volatile, large and positive correlation while the United States has the smallest correlation, the correlation of the United Kingdom and France follow almost the same trend and the values are a little bit larger than in the United States and a lot smaller than in Japan.

1.2 Literature Review

In recent years, some achievements have been made with ARMA model, ARCH model and GARCH model in the study of fluctuation in Chinese mainland stock market. Wang, Yu and Zhou (2014) made an empirical analysis of the relationship between the return rate and the risk of the Shanghai composite index. The results shows that there is a good fitting effect, which reflects the relationship between the return rate and the risk. This makes me not only study the return rate but also have volatility to measure the risk when studying the Chinese mainland stock market. Lu (2006) applied non-parametric GARCH model to predict the volatility of Chinese stock market better. He (2008) used ARIMA and ARCH models to predict stock prices and the results show that ARCH model is more satisfactory. Therefore, I combine ARMA model and GARCH model.

Engle (1982) proposed the Autoregressive Conditionally Heteroskedastic (ARCH) model to solve the problem of the volatility of inflation rate time series, this is the first theoretical model of volatility defined as conditional standard deviation. Bollerslev (1986) proposed Generalized Autoregressive Conditionally Heteroskedastic (GARCH) model specifically for financial data, which is especially suitable for the analysis and prediction of volatility. The Exponential-GARCH (EGARCH) model proposed by Nelson (1991) allows the positive and negative return on assets to have an asymmetric effect on volatility. The Threshold-GARCH (T-GARCH) model by Zakoian (1994) is another volatility model that reflects leverage. In order to study the portfolio of assets, the univariate GARCH model is extended to the multivariate GARCH model. Bollerslev (1990) first proposed Constant Conditional Correlation (CCC-) GARCH model, Engle and Sheppard (2001) extended CCC-GARCH model by setting the correlation matrix varied over time, hence Dynamic Conditional Correlation (DCC)- GARCH model is designed.

Solnik (1972) incorporated exchange rate and interest rate risks into the capital market-based asset-pricing model. Through the analysis of seven countries' stock markets, it is concluded that the principal factors affecting the country's capital price are still domestic factors, but other countries also have some influence on the country's capital price. On this basis, King and Wadhwani (1990) applied the model to the markets of many countries, further proving that the stock price of a country is influenced by the information of its own public information as well as the stock price information of other countries. Karolyi (1996) found that there is a two-way transmission between Japanese and the U.S stock markets. Hu and Xu (2003) found that the Hong Kong stock market has a stronger correlation with the volatility of Shanghai composite index than the U.S stock market. At the same time, they also found that the fitting degree of the regression coefficients of the volatility of the Shanghai composite index and the S&P index is extremely low, indicating that the correlation between the U.S and Chinese mainland is still extremely weak. Chen, Wu and Liu (2006) found that the linkage of stock markets in 11 countries and regions in the Asia-Pacific region is significantly enhanced after the Asian financial crisis. Chinese mainland stock market has a weak exogenous, except that there existed a short-term guiding relationship with the U.S stock market, there is no stable cointegration relationship between the Chinese mainland stock market and other stock markets. Zhang (2005) also proved that there is no long-term cointegration relationship between the international stock market and Chinese mainland stock market using E-GARCH model. However, Hong Kong, London and New York markets had one-way short-term spillovers on the Chinese mainland stock market mostly before the 1997 financial crisis.

2 Theory and Formulas

In this section, I will introduce ARMA model, ARMA-GARCH model, GARCH-in-Mean model, Exponential-GARCH Model, Dynamic Conditional Correlation-GARCH Model and Value-at-Risk method.

2.1 ARMA Model

AR model has truncated property of partial autocorrelation function and MA model has correlation function truncation property. The ARMA model combines the AR and MA models. a simpler model can be derived when the goodness of data fitting is similar, the partial autocorrelation function truncation and correlation function truncation are not required.

Autoregressive moving average model, ARMA(p, q) is defined to be

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{i=0}^q \theta_i \epsilon_{t-i} \quad (1)$$

where Y_t is the implied random variable, $\{\epsilon_{t-i}\}_{t=-\infty}^{\infty}$ is a white noise process. $c, \phi_i \in \mathbb{R}$ and $|\phi_1| < 1$. $\theta_0 = 1, |\theta_1| < 1$ and $\theta_i \in \mathbb{R}$. p represents the p th-order autoregressive process and q represents the q th-order moving average process.

Alternatively,

$$\phi(B)Y_t = \theta(B)\epsilon_t + c \quad (2)$$

where the lag operator $\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$, and $\theta(B) = 1 + \theta_1 B + \dots + \theta_q B^q$.

The stationarity of the ARMA(p, q) model, depends entirely on the stability of $\phi(B)$ not on $\theta(B)$, where p, q are finite.

The establishment of ARMA model can be tried from the lower order model one by one, the smaller the p+q the better. The model with the minimum Akaike information criterion is selected and the maximum likelihood method is used to estimate the parameters. White noise test is performed on the residuals to evaluate the adequacy of the model.

2.2 GARCH Model Family

In this study, ARMA-GARCH model is built for univariate analysis of Chinese mainland stock market, DCC-ARMA-GARCH model is built for multivariate analysis of worldwide stock market and Value-at-Risk is used for describe and forecast the volatility of market.

2.2.1 GARCH Model

Engle(1982) proposed Autoregressive Conditionally Heteroskedastic (ARCH) model to capture the volatility of financial market, with the condition of the disturbing sequence of stock returns $\epsilon_t = r_t - \mu$ are serially uncorrelated, but dependent.

ARCH(m) is defined to be

$$\begin{aligned}\epsilon_t &= \sigma_t z_t \\ \sigma_t^2 &= \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \dots + \alpha_m \epsilon_{t-m}^2\end{aligned}\quad (3)$$

where ϵ_t is the disturbing sequence of stock returns, $z_t \sim N(0, 1)$, σ_t^2 is the conditional variance, $\alpha_0 > 0$, and $\alpha_i \geq 0 (i = 1, 2 \dots m)$.

However, for typical financial time series, ARCH model needs to estimate a very large number of coefficients. In order to solve this problem, Bollerslev(1986) suggested the Generalized Autoregressive Conditionally Heteroskedastic (GARCH) model.

GARCH(m,s) is defined to be

$$\begin{aligned}\epsilon_t &= \sigma_t z_t \\ \sigma_t^2 &= \alpha_0 + \sum_{i=1}^m \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^s \beta_j \sigma_{t-j}^2\end{aligned}\quad (4)$$

where $z_t \sim N(0, 1)$, $\alpha_0 > 0$, $\alpha_i \geq 0 (i=1 \dots m)$, $\beta_j \geq 0 (j=1 \dots s)$ and $\alpha + \beta > 0$.

The most commonly used model is GARCH(1,1), which can be written as

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \quad (5)$$

2.2.2 ARMA-GARCH Model

If the mean equation is presented by ARMA model and GARCH model express conditional variance, then ARMA-GARCH can be written as

$$\begin{aligned}
r_t &= c + \sum_{i=1}^p \phi_i r_{t-i} + \sum_{i=0}^q \theta_i \epsilon_{t-i} \\
\epsilon_t &= \sigma_t z_t \\
\sigma_t^2 &= \alpha_0 + \sum_{i=1}^m \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^s \beta_j \sigma_{t-j}^2
\end{aligned} \tag{6}$$

Where r_t is the log returns of stocks and ϵ_t is the disturbing sequence of the log return of stocks.

The conditional variance σ_t^2 fluctuates as return change. Larger fluctuations are usually followed by larger fluctuations and smaller fluctuations follow smaller, which is called volatility clustering, this phenomenon stems from the persistent impact of external shocks on the volatility of stock price.

2.2.3 GARCH-in-Mean Model

Due to the relationship between risk and return, GARCH-in-Mean model is constructed to account for risk premium by letting conditional mean function depends on the contemporaneous conditional volatility.

The efficient market hypothesis (EMH) stating that there is no autocorrelation in r_t . However, r_t is autocorrelated in reality, because volatility is autocorrelated and r_t contains volatility.

GARCH(1,1)-M is defined to be

$$\begin{aligned}
r_t &= \mu + c\sigma_t^2 + \epsilon_t \\
\epsilon_t &= \sigma_t z_t \\
\sigma_t^2 &= \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2
\end{aligned} \tag{7}$$

where c in equation r_t is risk premium parameter, positive c implies return is positively correlated with volatility.

Serial correlation in r_t comes from serial correlation in σ_t^2 . The existence of risk premium is one of the sources of serial correlation in the return series.

2.2.4 Exponential-GARCH Model

Due to positive and negative return have the asymmetric impact on volatility, the fall is more drastic than the rise. Since GARCH model can not distinguish the positive and negative movements, the Exponential-GARCH model (E-GARCH(m,s)) proposed by Nelson(1991) is defined to be

$$\begin{aligned} z_t &= \sigma_t / \epsilon_t \\ \ln \sigma_t^2 &= \alpha_0 + \sum_{i=1}^s g(z_{t-i}) + \sum_{j=1}^m \beta_j \ln \sigma_{t-j}^2 \\ &= \alpha_0 + \sum_{i=0}^s (\alpha_i z_{t-i} + \gamma_i |z_{t-i}|) + \sum_{j=1}^m \beta_j \ln \sigma_{t-j}^2 \end{aligned} \quad (8)$$

where $g(z_t) = \alpha z_t + \gamma(|z_t| - E[|z_t|])$, $E(z_t) = 0$, α , β and γ are constants.

In E-GARCH model, $\ln \sigma_t^2$ serial depends on $\{\epsilon_t\}$ through $z_{t-j} = \epsilon_{t-j} / \sigma_{t-j}$, which can be used to describe the different effects of positive and negative return on volatility. But the σ_t^2 of GARCH model directly depends on ϵ_{t-j}^2 , so the positive and negative return have the same effect on volatility in GARCH model.

In this study, I use E-GARCH(1,1) model which is defined as

$$\ln \sigma_t^2 = \alpha_0 + \alpha z_{t-1} + \gamma |z_{t-1}| + \beta \ln \sigma_{t-1}^2 \quad (9)$$

2.2.5 Dynamic Conditional Correlation-GARCH Model

Unlike ordinary GARCH models, which can only capture and predict the static volatility of a time series, Dynamic Conditional Correlation (DCC)- GARCH model can analyze the dynamic fluctuations between several time series by decomposing conditional covariance matrix into conditional standard deviations and conditional correlation matrix. In other words, the fluctuation is a coefficient that changes with time.

DCC-GARCH model can be defined as

$$\begin{aligned}
r_t &= \mu_t + \epsilon_t \\
\epsilon_t &\sim N(0, \Sigma_t) \\
\Sigma_t &= D_t R_t D_t
\end{aligned} \tag{10}$$

Where Σ_t is the dynamic conditional covariance matrix, $D_t = \text{diag}(\sigma_1, \sigma_2 \dots \sigma_p)$ is dynamic conditional standard deviations matrix and R_t is dynamic conditional correlations matrix.

R_t needs to be positive to make sure Σ_t is positive, Engle (2002) introduced the following standardization

$$\begin{aligned}
R_t &= Q_t^{*- \frac{1}{2}} Q_t Q_t^{*- \frac{1}{2}} \\
Q_t &= (1 - \alpha - \beta)Q + \alpha a_{t-1} a_{t-1}' + \beta Q_{t-1} \\
a_t &= D_t^{-1} \epsilon_t
\end{aligned} \tag{11}$$

Where the diagonal matrix Q_t^* is made up of the diagonal elements of Q_t , Q_t drives the dynamics of the conditional correlations, a_t represents the standardized residuals and Q is the unconditional covariance matrix of a_t .

To ensure positive Σ_t , the two scalar parameters need to satisfy $\alpha > 0$, $\beta > 0$ and $\alpha + \beta < 1$.

Specifically, DCC(1,1)-GARCH(1,1) model can be written as

$$\begin{aligned}
r_t &= \mu_t + \epsilon_t \\
\epsilon_t &= \sigma_t z_t \sim N(0, \Sigma_t) \\
\Sigma_t &= D_t R_t D_t \\
D_t &= \text{diag}(\sigma_1, \sigma_2 \dots \sigma_p) \\
\sigma_t^2 &= \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\
R_t &= Q_t^{*- \frac{1}{2}} Q_t Q_t^{*- \frac{1}{2}} \\
Q_t &= (1 - \alpha - \beta)Q + \alpha a_{t-1} a_{t-1}' + \beta Q_{t-1} \\
a_t &= D_t^{-1} \epsilon_t
\end{aligned} \tag{12}$$

2.3 Value-at-Risk Method

Because of some extreme price movements, Value-at-Risk (VaR) is used to figure out the maximal loss of a financial position during given period for a given probability.

For a long position, $VaR_t(p, h)$ is defined as

$$F_h(VaR_t(p, h)) = Prob[\Delta V(h) \leq VaR_t(p, h)] \quad (13)$$

For a short position, $VaR_t(p, h)$ is defined as

$$1 - F_h(VaR_t(p, h)) = Prob[\Delta V(h) \geq VaR_t(p, h)] \quad (14)$$

Assume t is current time and h is given period, $\Delta V(h)$ is the change in the value of the assets in the position from t to $t+h$, and $F_h(x)$ is the cumulative distribution function of $\Delta V(h)$.

Define the p th quantile of $F_h(x)$ as

$$x_p = \inf\{x \mid F_h(x) \geq p\} \quad (15)$$

Hence, $VaR_t(p, h) = x_p$.

3 Empirical studies

3.1 Data

All the data downloaded from a stock software, provided by Nuclear New Flush Network Information Co. LTD, called Straight flush. I consider the daily values of Shanghai (securities) composite index from December 19, 1990 to February 28, 2020. The Shanghai composite index is a statistical index reflecting the overall trend of listed stocks on the Shanghai stock exchange, which can measure Chinese mainland stock market and is not openly traded in the stock market.

In order to study the impact of other financial markets in the world on the mainland stock market, I choose other 5 indexes from December 1990 to February 2020. The s&p 500, which covers 500 of the most popular stocks, represents the United States market; The FTSE 100 index covers the 100 most valuable stocks in the London stock market; The nikkei 225 is Japan's

representative stock index, a barometer of whether the country's stock or capital markets are good or not and one of the factors that reflects the value of the yen; The hang seng index is the most influential stock price index in Hong-Kong, which is composed of 33 representative listed stocks in the Hong-Kong stock market; The CAC40 index is made up of 40 French stocks and reflects price fluctuations in the French stock market.

The research process focuses on the logarithmic return rate of stocks rather than the stock price, because the return rate can satisfy the Weak stationary requirements for time series modeling. Let P_t be the daily price, then the log return rate of stock is defined as

$$r_t = \ln(P_t) - \ln(P_{t-1})$$

3.2 Univariate Analysis

The univariate analysis part is focus on the study of the return rate and risk of Chinese mainland stock market. At first, it can be seen from the upper plot in figure 1 that the return rate fluctuates around the value of zero, therefore I can preliminary judge that the daily return of Shanghai composite index is stable. It can also be seen that the fluctuation of the return has peaks and troughs, the high fluctuation follows the high fluctuation and the low fluctuation follows the low fluctuation, which indicates volatility clustering appears.

3.2.1 ARMA-GARCH

Since the ARMA model is modeled for stationary sequences, I need to further examine the stationarity of the sequence. The autocorrelation function (ACF) and the partial autocorrelation function (PACF), also shown in figure 1, suggest that the sequence of returns is weakly stationary and random due to little serial correlation.

Augmented Dickey-Fuller (ADF) test and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test are both unit root tests, which are used to check for stationarity. The p-value of ADF test is smaller than 0.01 and the P-value of KPSS test is lager than 0.05, which both show that the return series is stable. The p-value of Ljung-Box Test in lag 12 and lag 24 are both far less than 0.05, which indicate that the return series is stationary non-white noise sequence and can fit ARMA model.

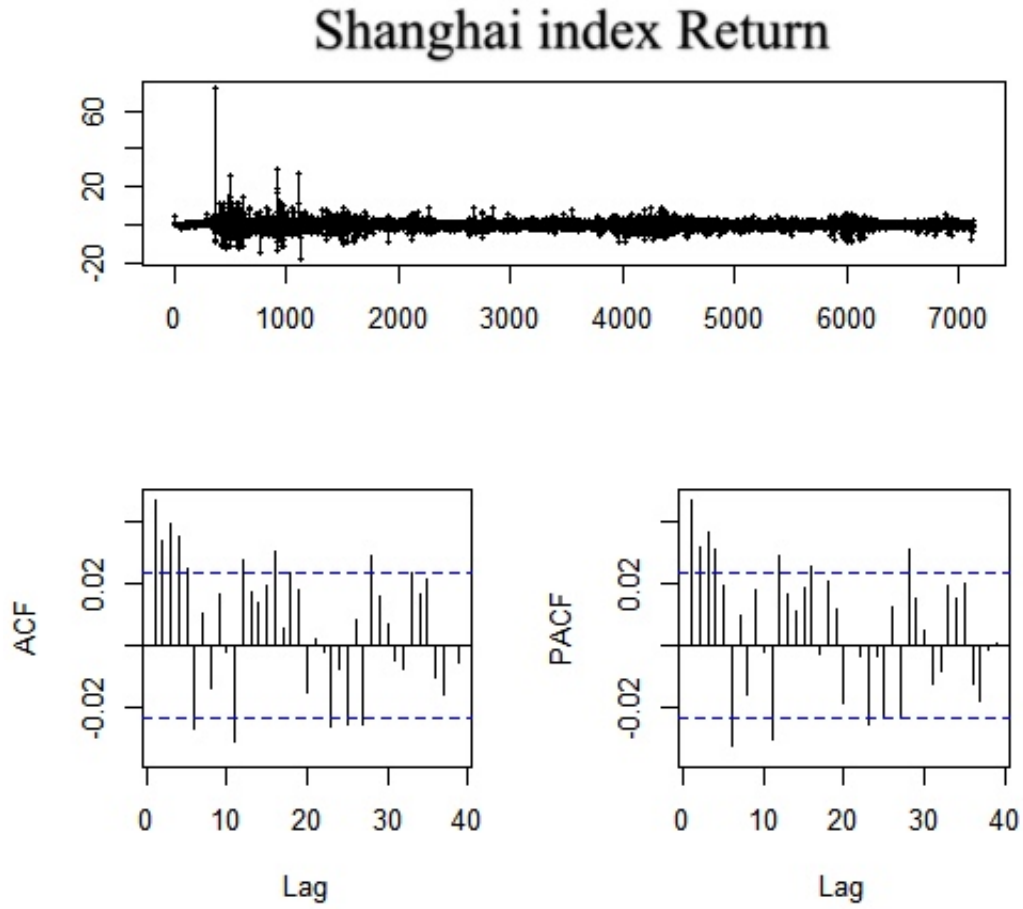


Figure 1: Log Return of Chinese mainland

In order to determine the order of the model, I not only follow the truncation of ACF and PACE, but also combine the principle of AIC and BIC minimization. Hence I get ARMA(3,3) model, which can be written as

$$r_t = 0.8368a_{t-1} + 0.5035a_{t-2} - 0.6878a_{t-3} + \epsilon_t - 0.797\epsilon_{t-1} - 0.5119\epsilon_{t-2} + 0.6873\epsilon_{t-3} \quad (16)$$

0.1215
0.1733
0.0964
0.1199
0.1639
0.0914

The standard errors of the coefficients are 0.1215, 0.1733, 0.0964, 0.1199, 0.1639 and 0.0914, which show that all the coefficients are significant at the significance level of 0.05. Ljung-Box Test is used to test the residual terms of the model, the p-value in lag 6 and lag 8 are both larger than 0.05, which prove that the residual sequence of each model has no significant correlation. Combined with figure 11 in appendix A, all show that the fitted ARMA(3,3) model is sufficient and reasonable.

By observing the sequence diagram of the return rate, it is found that the fluctuation range of the return is not fixed, with heteroscedasticity and aggregation. In order to further verify, I am going to test the squared residual term of ARMA(3,3) model.

From figure 12 in appendix I can see that ϵ_t^2 exists serial correlation. Adopting Ljung-Box test for squared residuals, the P-value is smaller than 0.05 and close to 0, the p-value of Lagrange multiplier test is also close to 0, which all confirm that the residual term of ARMA(3,3) model has a strong ARCH effect.

Therefore, I am going to build GARCH(1,1) model. The skewness of residual term is 5.246 and the kurtosis is 156.116, which indicate that the residual term follow a heavy-tailed distribution. And the p-value of Jarque Bera Test close to 0, means the residual term does not follow normal distribution.

I plan to build ARMA(3,3)-GARCH(1,1) model through rugarch package. First I suppose z_t follows normal distribution,

$$\begin{aligned} r_t &= -0.217r_{t-1} - 0.395r_{t-2} - 0.801r_{t-3} + \epsilon_t + 0.227\epsilon_{t-1} + 0.37\epsilon_{t-2} + 0.829\epsilon_{t-3} \\ \epsilon_t &= \sigma_t z_t, z_t \sim N(0, 1) \\ \sigma_t^2 &= 0.053 + 0.147\epsilon_{t-1}^2 + 0.852\sigma_{t-1}^2 \end{aligned} \quad (17)$$

All the coefficients except α_0 are highly significant at the significance level of 0.05 and the statistics of other model tests show that the model fitting is sufficient. The log-likelihood is -13676.28, the AIC of the model is 3.836 and BIC is 3.845.

Due to the heavy tailed of residual term caused by the huge value of kurtosis, then I let z_t follows student-t distribution,

$$\begin{aligned} r_t &= -0.42r_{t-1} + 0.842r_{t-2} + 0.54r_{t-3} + \epsilon_t + 0.454\epsilon_{t-1} - 0.811\epsilon_{t-2} - 0.512\epsilon_{t-3} \\ \epsilon_t &= \sigma_t z_t, z_t \sim t^*(4.158) \\ \sigma_t^2 &= 0.029 + 0.106\epsilon_{t-1}^2 + 0.893\sigma_{t-1}^2 \end{aligned} \quad (18)$$

All the coefficients are highly significant at the significance level of 0.05 and z_t follows student-t distribution with a degree of freedom of 4.158. The log-likelihood is -12856.62, the AIC of

the model is 3.607 and BIC is 3.616.

Since the skewness larger than 0, there exists a right skewed distribution, finally I let z_t follows skewed-student-t distribution,

$$\begin{aligned} r_t &= -0.406r_{t-1} + 0.835r_{t-2} + 0.531r_{t-3} + \epsilon_t + 0.439\epsilon_{t-1} - 0.806\epsilon_{t-2} - 0.502\epsilon_{t-3} \\ \epsilon_t &= \sigma_t z_t, z_t \sim t^*(0.96, 4.161) \\ \sigma_t^2 &= 0.03 + 0.108\epsilon_{t-1}^2 + 0.891\sigma_{t-1}^2 \end{aligned} \quad (19)$$

All the coefficients are highly significant at the significance level of 0.05, the log-likelihood is -12852.4, the AIC of the model is 3.605 and BIC is 3.616.

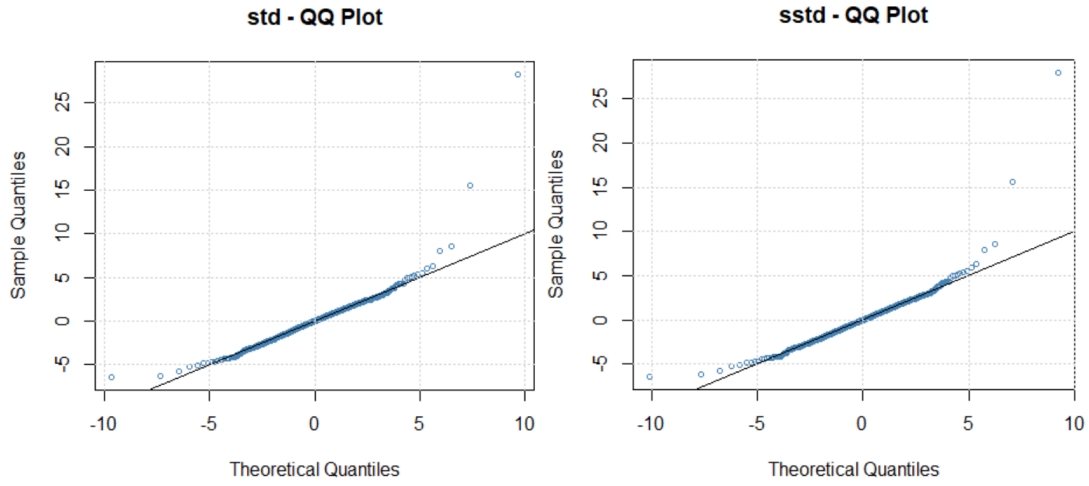


Figure 2: QQ Plot

The QQ plots given in figure 2 show that the residuals in two distributions are basically a straight line except for a few deviating. I choose the one with skewed-student-t distribution as my final ARMA-GARCH model, since it has the largest log-likelihood value and the smallest AIC and BIC, which is equation(19).

3.2.2 ARMA-GARCH-M

Due to the return on a financial asset may depend on its volatility, I add variance into mean equation and build one ARMA-GARCH-M model with skewed-student-t distribution.

$$\begin{aligned} r_t &= 1.104 + 1.109r_{t-1} + 0.657r_{t-2} - 0.766r_{t-3} + \epsilon_t - 1.06\epsilon_{t-1} - 0.693\epsilon_{t-2} - 0.756\epsilon_{t-3} + 0.013\sigma_t^2 \\ \epsilon_t &= \sigma_t z_t, z_t \sim t^*(0.97, 4.171) \\ \sigma_t^2 &= 0.029 + 0.108\epsilon_{t-1}^2 + 0.891\sigma_{t-1}^2 \end{aligned} \quad (20)$$

All the coefficients are highly significant at the significance level of 0.05, the log-likelihood is -12840.73, the AIC of the model is 3.603 and BIC is 3.616.

3.2.3 ARMA-E-GARCH

In stock market, the distribution of returns not only has the characteristic of peak and heavy tail, but also has the asymmetric effect on the fluctuation of returns. To solve these, I build one ARMA-E-GARCH model.

$$\begin{aligned} r_t &= -0.354r_{t-1} + 0.816r_{t-2} + 0.47r_{t-3} + \epsilon_t + 0.393\epsilon_{t-1} - 0.781\epsilon_{t-2} - 0.438\epsilon_{t-3} \\ \epsilon_t &= \sigma_t z_t, z_t \sim t^*(0.96, 3.997) \\ \ln \sigma_t^2 &= 0.015 - 0.059(|z_{t-1}| + 0.228z_{t-1}) + 0.983 \ln \sigma_{t-1}^2 \end{aligned} \quad (21)$$

All the coefficients are highly significant at the significance level of 0.05, the log-likelihood is -12796.79, the AIC of the model is 3.59 and BIC is 3.602.

3.2.4 Value-at-Risk

Based on the principle that the higher the log-likelihood, the smaller the AIC and BIC, the better the model. From table 1, it is obviously that ARMA-E-GARCH model is the best one, shows that the leverage effect is significant in the Shanghai composite index return and the model fitting is significantly improved by adding the asymmetric effect parameters. Meanwhile, ARMA-GARCH-M model is only a little bit better than ARMA-GARCH model, proves that the correlation between volatility and return rate of Chinese mainland stock market is not so obvious, that is, high risks do not automatically produce high returns.

Due to the value of AIC and BIC of the ARMA-E-GARCH models follow skewed-student-t distribution and student-t distribution are basically the same, I replace skewed-student-t by

	ARMA-GARCH	ARMA-GARCH-M	ARMA-E-GARCH
log-likelihood	-12852.4	-12840.73	-12796.79
AIC	3.605	3.603	3.59
BIC	3.616	3.616	3.602

Table 1: Information Criteria of Models

student-t distribution for calculating quantile easily. Hence I get new ARMA(3,3)-E-GARCH(1,1) model.

$$\begin{aligned}
r_t &= -0.369r_{t-1} + 0.822r_{t-2} + 0.479r_{t-3} + \epsilon_t + 0.407\epsilon_{t-1} - 0.786\epsilon_{t-2} - 0.448\epsilon_{t-3} \\
\epsilon_t &= \sigma_t z_t, z_t \sim t^*(3.994) \\
\ln \sigma_t^2 &= 0.015 - 0.059(|z_{t-1}| + 0.225z_{t-1}) + 0.984 \ln \sigma_{t-1}^2
\end{aligned} \tag{22}$$

I obtain conditional mean and conditional standard deviation from the model above, draw Q-Q plot to check the standardized residuals z_t against their specified t in figure 13 in appendix.

I consider the Value-at-Risk confidence level as 0.99 and 0.95, which equals to the probability is 0.01 and 0.05. Then I calculate VaR in two ways, one is using the equation $VaR = \mu + \sigma * q(p)$, with $\sigma^2 = (n/n - 2)v^2$, where μ , n and v computed from E-GARCH model. The other one is replace $\sqrt{n/n - 2} * q(p)$ as qdist function in rugarch package.

At last I get the same results and I draw the plot of return rate and Value-at-Risk which can be seen in figure 3, the actual return rate is the dark gray line, the calculated VaR is represented by red with 5% VaR limit and the other 1% VaR is represented by blue.

After that I perform a rolling approach to do one-step ahead estimation and forecasting of the ARMA-E-GARCH model consisted of Shanghai index return.

In general, one-third of the data will be selected for forecasting. Since the return of Shanghai index has 7134 observations, then the first 5000 observations are chosen as estimating data and the last 2134 as the test set, which is also called out-of-sample data. Since the data contains daily return, I set the refit window size for 2134 forecasts and one day frequency. I calculate 5%

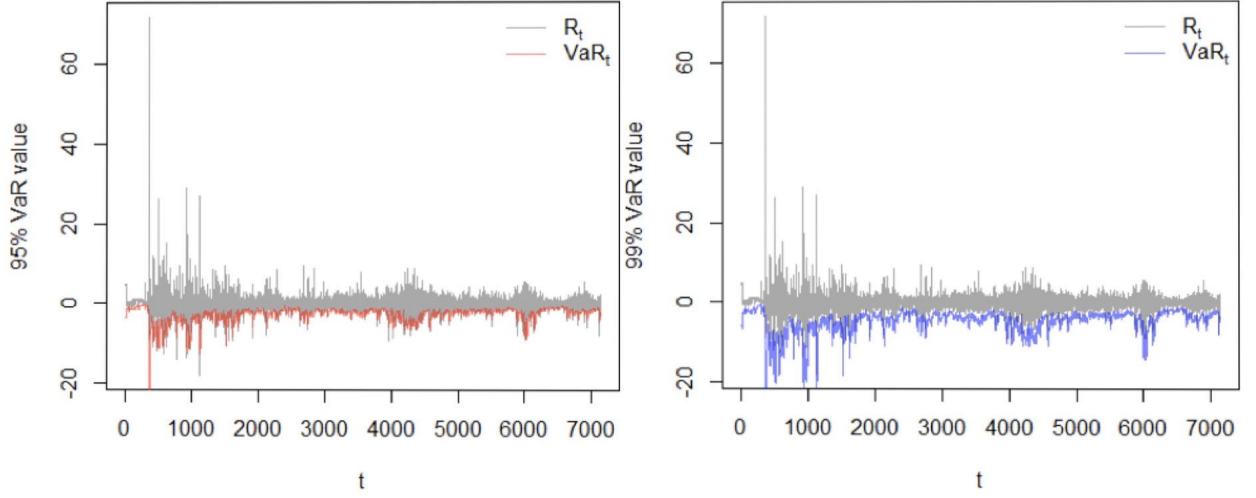


Figure 3: Return rate and VaR

and 1% VaR using ugarchroll function, VaR backtest forecasts with two different confidence level against the actual return can be plotted in figure 4 below.

Same as figure 3, the dark gray points in figure 4 represent the actual return rate, the black line represents the calculated VaR and the red dots lead to lower return rate than VaR, which are also the number of exceedances.

It can be seen from figure 4 that although the VaR for $\alpha = 0.01$ and $\alpha = 0.05$ apparently following the same trend, the value of VaR for 0.01 is smaller than the value for 0.05, that is VaR for $\alpha = 0.05$ is located upper than another, which leads to more exceedances. And it is obviously that at 95% confidence level VaR produces more exceedances since there are so many conspicuous red dots. Some exceedances tend to be quite clustered in figure 4, which perhaps indicates that these two models haven't completely capture return rate's conditional volatility.

To test the success rate of VaR, I check the VaR exceedance by Kupiec test and Christoffersen test, perform the results of backtesting of ARMA(3,3)-E-GARCH(1,1) model in the table 2 as follow.

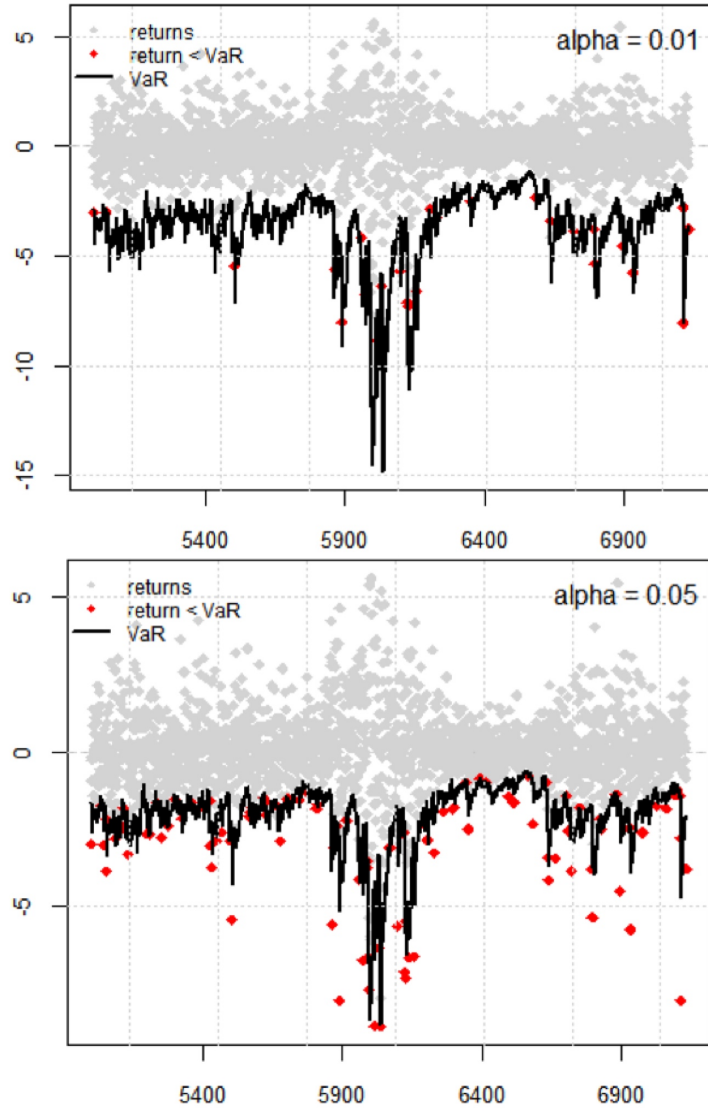


Figure 4: ARMA-E-GARCH model rolling forecast at 99% and 95% confidence level

The actual failures of $\alpha = 1\%$ are greater than the expected exceed, then this model is considered as underestimation while the actual exceeds of $\alpha = 5\%$ are a little bit smaller than the expected exceed, which is overestimated. However, it can be informed from table 2 that both unconditional and conditional L.R. statistics are smaller than critical value, which show that both Kupiec test and Christoffersen test don't reject null hypothesis on both 1% and 5%.

The results indicate that the exceedances of Value-at-Risk of ARMA(3,3)-E-GARCH(1,1) model for $\alpha = 0.01$ and $\alpha = 0.05$ is held at the expected number and independently. They also indicate that this ARMA-E-GARCH model can produce accurate VaR forecasts and han-

Alpha	1%	5%
Backtest Length	2134	2134
Expected Exceed	21.3	106.7
Actual VaR Exceed	27	105
Actual %	1.3%	4.9%
Unconditional Coverage	(Kupiec test)	
Null-Hypothesis:	Correct Exceedances	
LR.uc Statistic:	1.399	0.029
LR.uc Critical:	6.635	3.841
LR.uc p-value:	0.237	0.866
Reject Null:	NO	NO
Conditional Coverage	(Christoffersen test)	
Null-Hypothesis:	Correct Exceedances and Independence of Failures	
LR.cc Statistic:	2.368	0.296
LR.cc Critical:	9.21	5.991
LR.cc p-value:	0.306	0.862
Reject Null:	NO	NO

Table 2: VaR Backtest Report

de the ever-changing fluctuations in the return rate of Shanghai index well.

3.3 Multivariate Analysis

3.3.1 Data description

I choose the representative indexes of six countries or regions to establish a multivariate model, to study the impact of stock market fluctuations in five countries or regions including Hong-Kong, the United States, Japan, the United Kingdom and France on the volatility of Chinese mainland stock market. As the stock markets in different countries have different suspension dates, I delete the data that do not overlap in the trading days of several markets.

Firstly I perform a descriptive statistics on these six index returns, which can be seen in the table 3 below.

	Chinese	Hong-Kong	U.S	Japan	U.K	France
Min	-17.91	-14.73	-9.47	-12.11	-9.27	-9.47
Max	71.92	17.25	10.96	9.49	9.38	10.59
Mean	0.047	0.029	0.025	0.0005	0.0137	0.017
Median	0.07	0.06	0.04	0.03	0.04	0.04
Skew	5.314	-0.015	-0.205	-0.447	-0.154	-0.079
Kurtosis	157.69	9.502	8.979	5.061	6.418	4.878

Table 3: Descriptive statistics of six index

Table 3 shows that the log return of Chinese mainland stock market has the largest maximum value and the smallest minimum value, which perhaps indicates Chinese stock market has the largest volatility among these six stock market. All of these six indexes have positive daily return and the value of their mean are all close to 0.

All of these returns do not follow normal distribution due to their skew and kurtosis. But from table 3 above, I can see that Chinese return is different from the others, since it is positively skewed while the value of the other five are all slightly negative. They all have large kurtosis which indicate fat-tailed and non-symmetric leptokurtic. Moreover the return of Chinese index has the biggest skew and kurtosis, which also may indicate Chinese mainland stock market has large volatility and some big outliers.

In order to performance volatility more specifically and obviously, I draw the plot of these six time series in figure 5, where the x-axis is labelled by the number of daily return in dataset, not the ordinary dates. It can be seen that all of these six indexes return fluctuate around 0 and there exist volatility clustering. I can see that Chinese mainland stock market has a huge outlier at May 21,1992 and other five stock markets have the same obvious huge outliers at October 2008 due to the financial crisis.

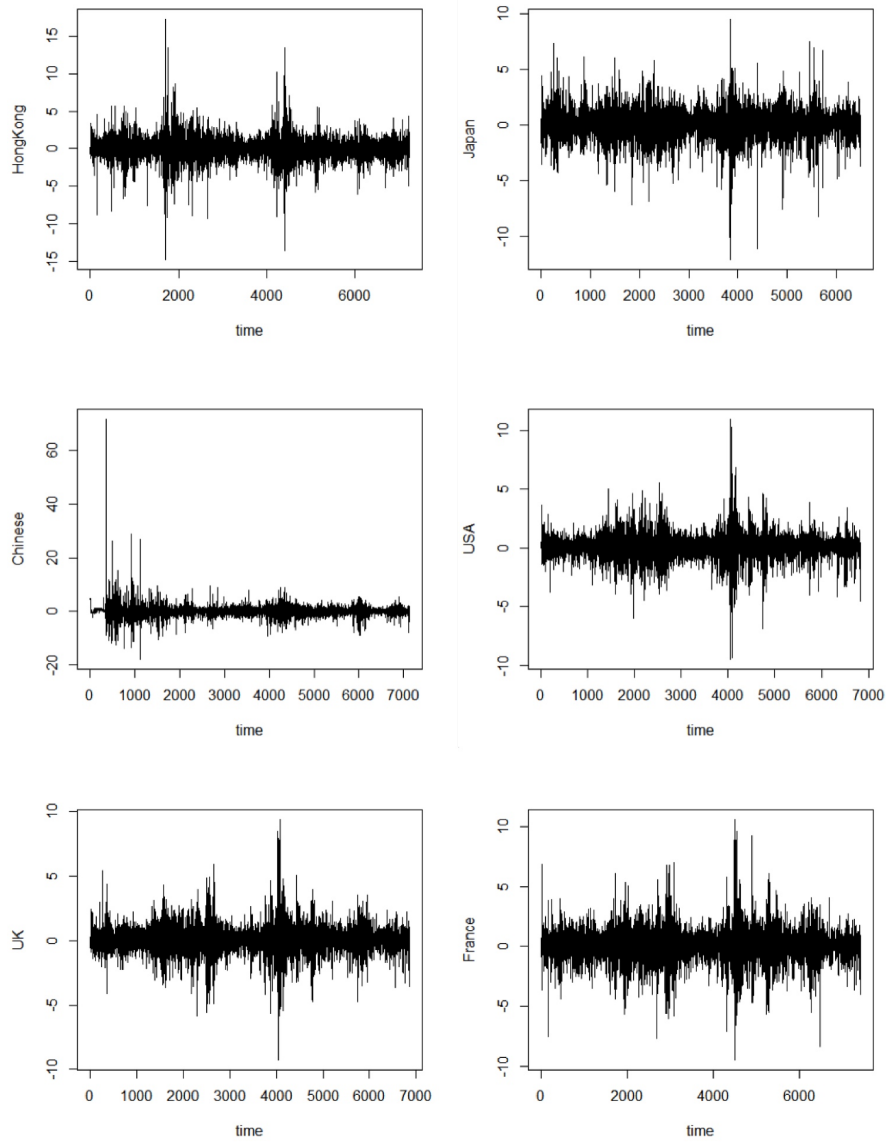


Figure 5: Worldwide stock return

At the same time, I also decide to build ARMA model of six return rate by choosing the minimum AIC and BIC, for better DCC-ARMA-GARCH model in the next step. These six ARMA models can be seen in table 5 in appendix, I may say that ARMA (3,3) is the mean model of the index return in most countries' stock markets.

3.3.2 DCC-ARMA-E-GARCH

To further study the influence of the stock markets of five different countries or regions on the Chinese mainland stock market, I decide to establish five DCC-GARCH models respectively. Due to the leverage effect and the results of model comparison in univariate study,

I choose E-GARCH model. Since ARMA-E-GARCH model with student-t distribution and with skewed-student-t distribution performs the same, I choose to follow the student-t distribution for easily study. I create five portfolios, each of them consists of the return of mainland China and another country or region, set the vector of weights on two indexes is (0.5,0.5).

	Hong-Kong	U.S	Japan	U.K	France
α	0.0427	0.0012	0.0054	0.0022	0.0025
β	0.9414	0.9984	0.9945	0.9976	0.9971
$\alpha + \beta$	0.9841	0.9996	0.9999	0.9998	0.9996

Table 4: Dynamic coefficients of DCC-GARCH models

The p-value of coefficients are all less than 0.05, which means they all have significant effects. As described by the DCC-GARCH model's definition, the scalars α and β must be positive, but at the same time their sum must be strictly smaller than one. Strong β means the persistence of conditional variance, then these five conditional variances seem to be persistent from table 4. The larger the α , the greater the sensitivity of markets to shocks but the α in these five models are relatively small.

The closer the sum gets to 1, the stronger the dynamic correlation between the two stock markets. Hence we can see that there exists the strongest volatility between Japan and Chinese mainland stock market, the correlation of the U.K is just a little bit smaller than Japan, the U.S and France have the dynamic correlation of the same intensity with mainland China while Hong-Kong's correlation is relatively most stable.

In these figures of the dynamic conditional correlation between five different countries or regions and Chinese mainland stock market, the general opinion is that over this twenty-year period the correlation tends to be upward trending. The correlations were negative a few times particularly at the start of the estimated period. In addition, the figures reveal correlations drop during economic crisis.

1. HongKong

From the upper plot in figure 6, it can be seen that the time-varying correlation between Chinese mainland stock market and Hong-Kong's fluctuates around zero. From the beginning it fluctuates between plus or minus 0.4 and later grows to 0.6, which informs that Hong-Kong stock market is becoming more relevant to the Chinese mainland market. The value of correlation coefficients are the largest among five international stock markets, the almost positive value in the middle and later periods both show that there is a positive and large correlation between two markets.

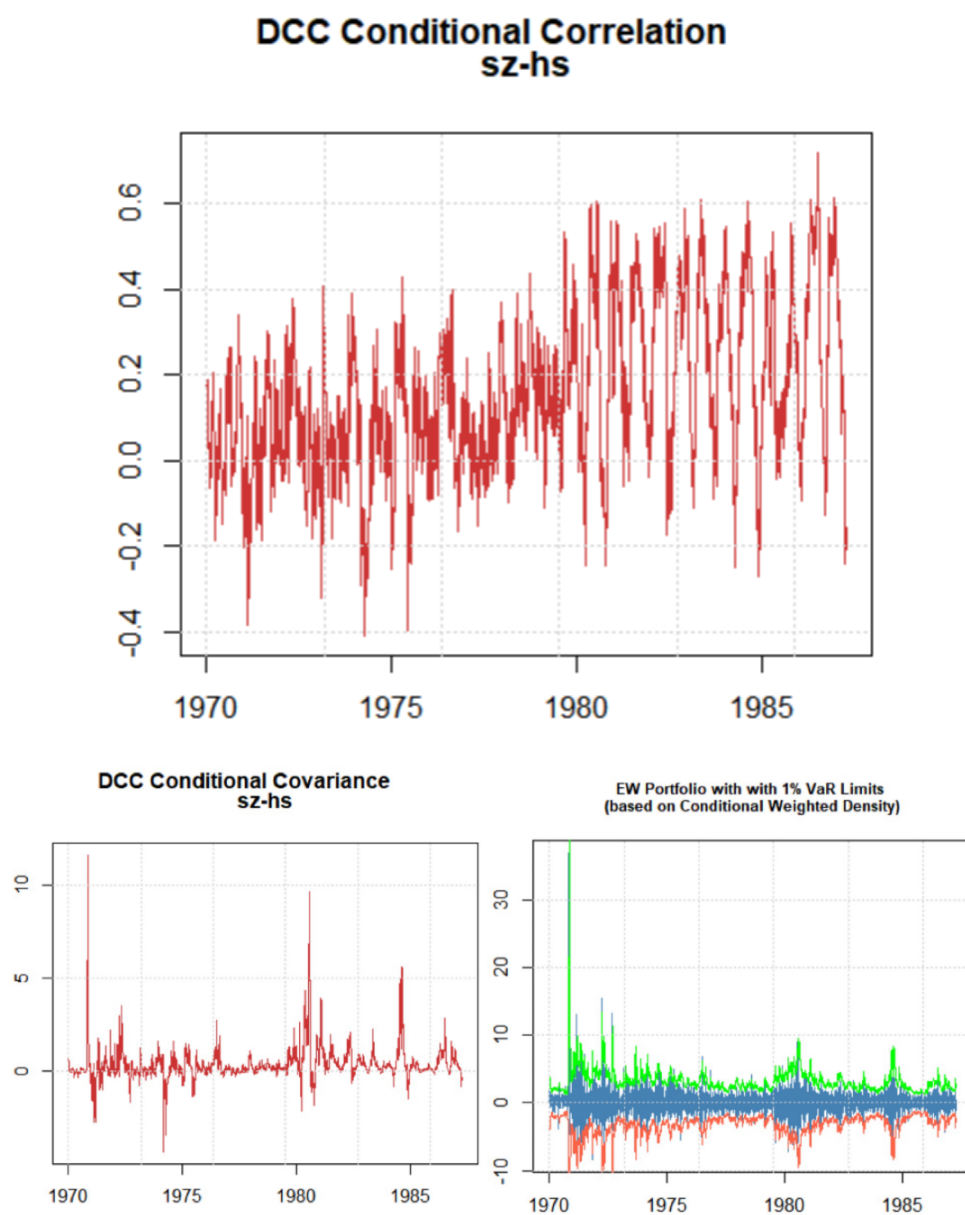


Figure 6: Between Chinese mainland and HongKong

The conditional covariance between Hong-Kong and Chinese mainland stock market is relatively stable, although the financial turmoil has caused several violent fluctuations, it does not affect the overall stable trend. The covariance fluctuates around zero but it's positive most of the time. The 1% VaR limits seems cover most of the returns of portfolio, except for a few exceedances when returns are volatile, especially at the beginning.

2. the United States

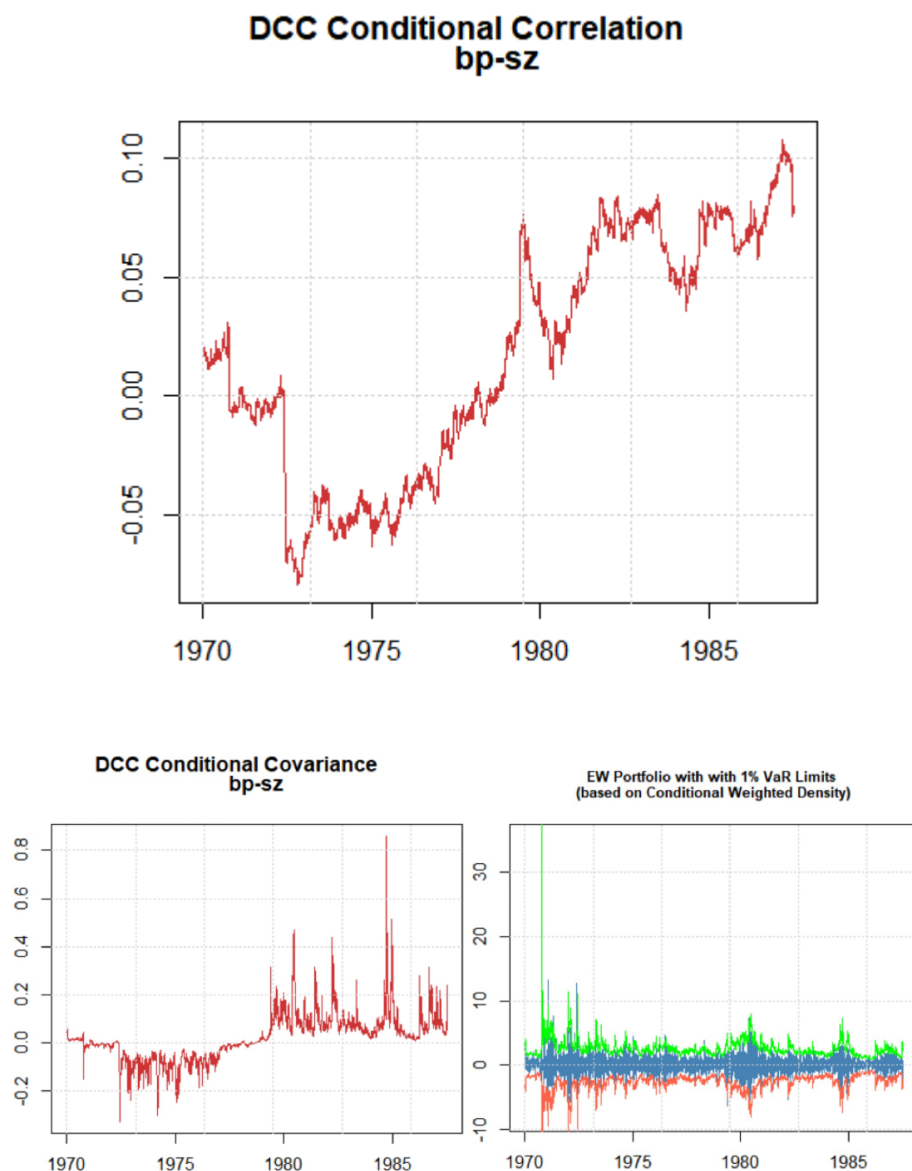


Figure 7: Between Chinese mainland and USA

Figure 7 shows the dynamic correlation between Chinese mainland's stock market and the U.S

starts around zero, after a sharp decrease the correlation is negative but rising from trough which is less than -0.05. In the medium period the correlation switches to positive, fluctuates around 0.07 and rises to 0.1 in the recent periods, which informs that the correlation is positive and continues to grow, but it still the smallest coefficients among five portfolios.

It can be informed that the conditional covariance between the U.S and Chinese mainland fluctuates around zero, fluctuates in negative values in the first decade and fluctuates in positive values later. Covariance fluctuates more frequently in the U.S than in any of the other four countries, with peaks and troughs throughout the trend. The 1% VaR limits still cover most of the returns of portfolio with a few exceedances.

3. Japan

As can be seen from figure 8, the correlation between Japan and Chinese mainland stock market shows a positive general trend. Although it fluctuates around zero in the first half, it's completely positive after that and it keeps larger value than other model's correlation, peaks at 0.5 at the end, which are the second large correlation coefficients. This seems to reflect the same thing as the coefficients in table 3, which is that Japan has large and positive correlation that fluctuates most frequently with the Chinese mainland stock market.

Same as above, the conditional covariance between Japan and Chinese mainland is also quite stable, almost all of them are positive and fluctuate between 0 and 2. The volatility in the first half of period is quite small, after a sudden increase caused by a financial shock, the conditional covariance returns to fluctuate near zero. Portfolio with 1% VaR still perform the same as above.

4. the United Kingdom

Figure 9 shows that the overall trend in the correlation between China and the U.K is similar to that between China and Japan. It also fluctuates around zero in the first half and totally positive in the back half. However, the conditional correlation in the U.K does not fluctuate as frequently as in Japan, it is mostly negative in the first half while Japan is half positive and half negative during this period. In the first half of the correlation, the fluctuation continues to decline until the trough of -0.1 is touched, then it rise back to 0 and quickly increase to 0.1,

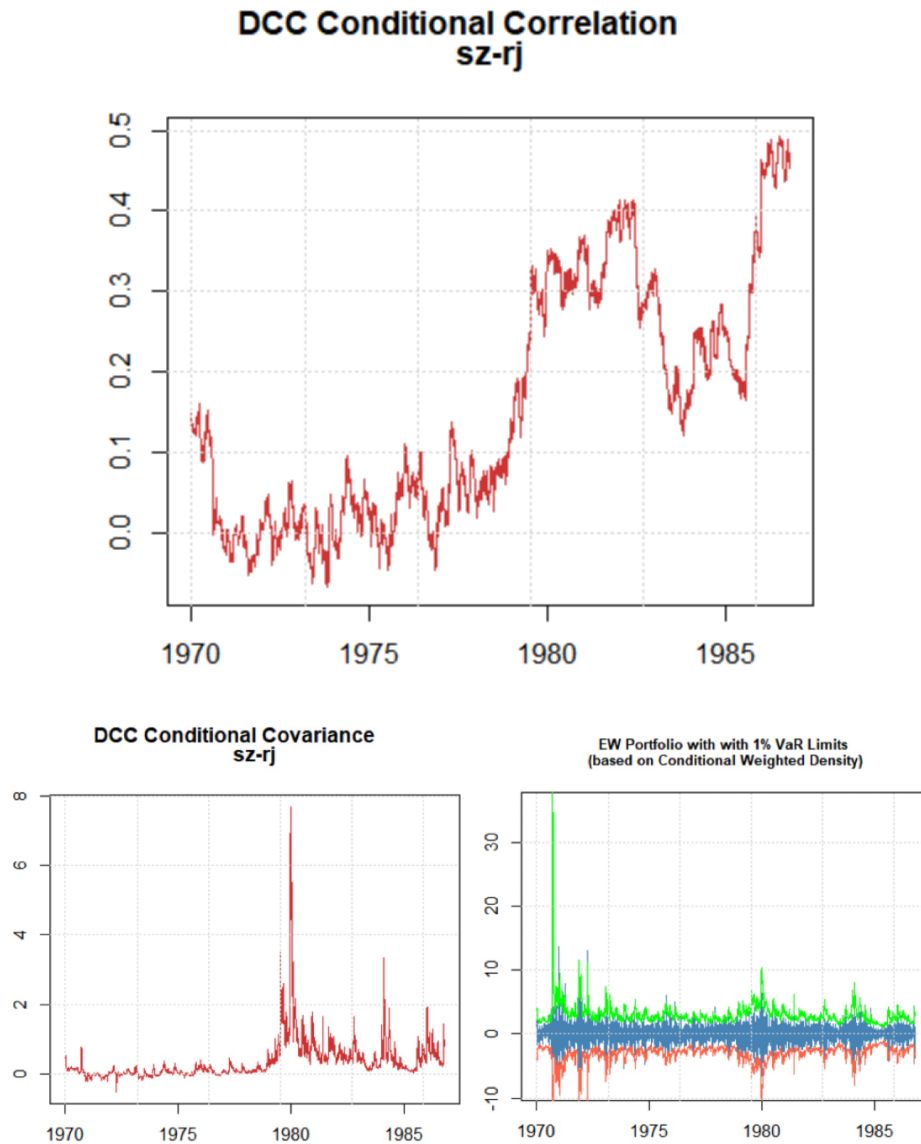


Figure 8: Between Chinese mainland and Japan

keep fluctuating around 0.2.

The overall trend of the conditional covariance between the U.K and Chinese mainland stock market is kind of similar with Japan above. But it fluctuates around zero in the first half of period with half positive and half negative. The trend of fluctuation in the next half is roughly the same as Japan while the second peak of the U.K is higher than Japan's.

5. France

Figure 14 in appendix indicates that the dynamic correlation coefficient between Chinese mainland stock market and France also shows an overall trend of decreasing first and then rising, but

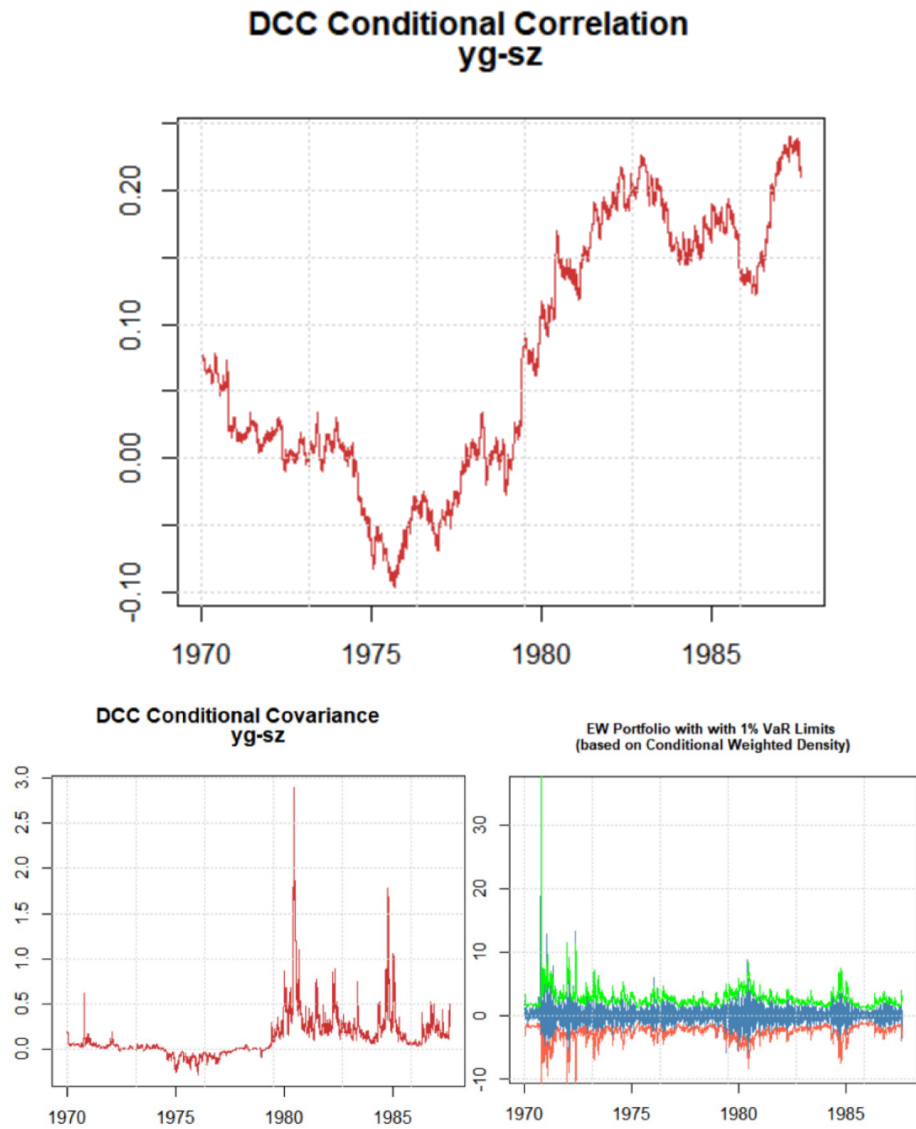


Figure 9: Between Chinese mainland and U.K

the correlation coefficient is positive for three quarters of the time. It has a positive correlation at the beginning, then switches to negative and fluctuates around -0.05 for a while, quickly increase to 0.1. The trend of correlation coefficients between China and France in the second half is almost the same as that in the U.K.

It is obviously that the trend of conditional covariance between France and Chinese mainland is almost the same as that between the U.K and Chinese mainland. The vast majority of covariance is positive and there exists two distinct peak in the second half of period. The 1% VaR limits still cover almost all of the returns of portfolio, except for a few exceedances, which may tell that the volatility of this portfolio seldom exceed expectation of Value-at-Risk, same

as other four portfolios.

3.3.3 Forecast

For further analysis, I forecast five DCC-GARCH models in 100 horizon. As can be seen from figure 10, the covariances between these five countries or region are all positive, only the covariance between Hong-Kong and Chinese mainland stock market is predicted to increase, while the other four countries have sharp downward trends. However, even if Hong Kong is

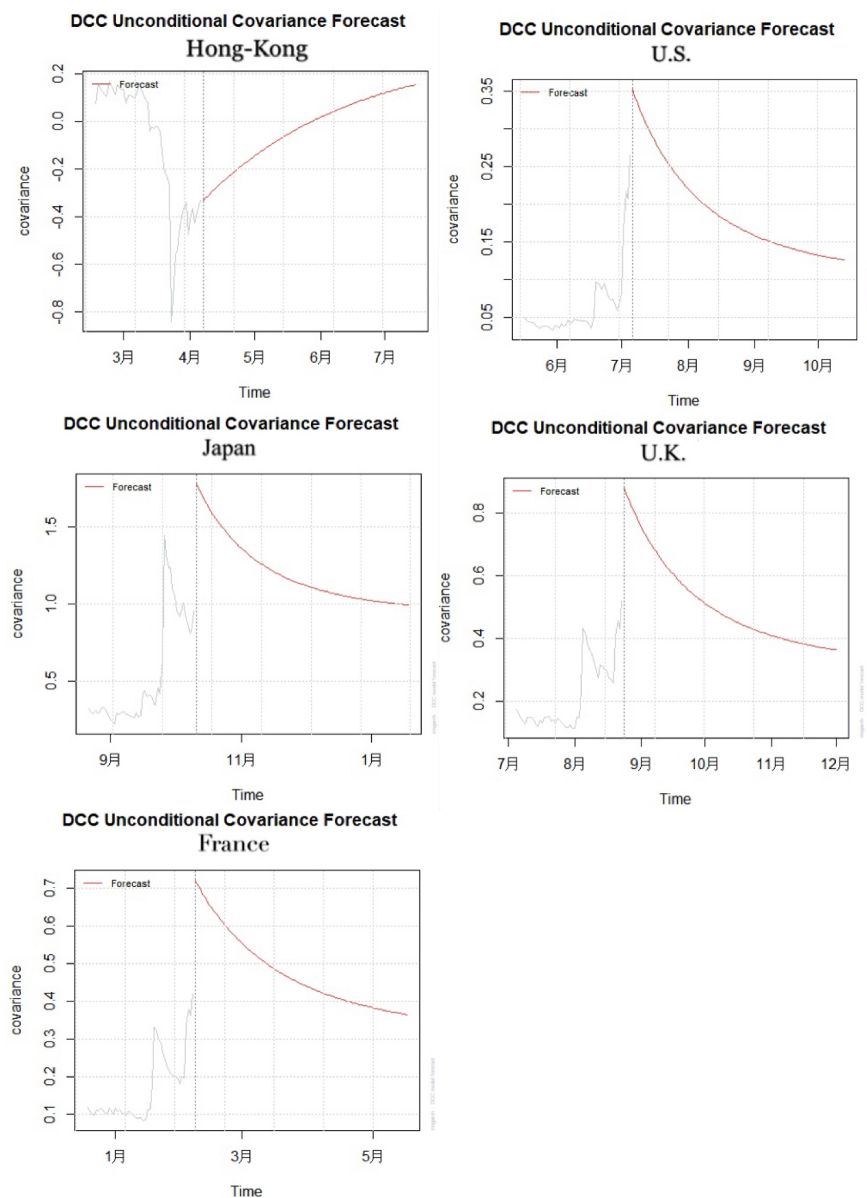


Figure 10: Unconditional covariance forecast

on the rise, its covariance forecast value is not as high as that of the other three countries, only exceeds that of the United States after the decline. Japan has the highest predicted covariance,

even after the decrease, while the covariance of the U.K. and France is not only the same downward trend but the predicted value is also within the same range.

3.3.4 Result

With its highest peaks and lowest troughs, Chinese mainland stock market has a wider range of fluctuations than the other five stock markets in the world.

Among five chosen foreign stock markets, the correlation between Hong-Kong and Chinese mainland stock market is relatively the most stable and largest. There exists the most volatile correlation between Japan and mainland China, though they are the second highest correlated. The correlation of the United Kingdom and France are similar in size and volatility, while the correlation between the United States and Chinese mainland stock market is surprisingly minimal with high volatility.

4 Conclusion

As an essential part of the financial market, Chinese mainland stock market not only carries out funding and resource allocation, but also performs major economic forecasting activities as the "barometer" of China's macroeconomic development.

With consideration of the significant heteroscedasticity, risk premium and leverage effect of the return rate of Chinese mainland stock market, hence ARMA(3,3)-E-GARCH(1,1) model follows skewed-t-distribution can generate better results, which may be the most fully fitted for the logarithmic return rate of Shanghai composite index. The risk premium parameter of ARMA(3,3)-GARCH(1,1)-M is 0.013, which indicates if the risk of return rate increases by one unit, the corresponding return rate will increase by 0.013.

The E-ARMA(3,3)-GARCH(1,1) model follows student-t-distribution performs quite well in forecasting the risk of daily return rate with the 99% and 95% confidence level. The backtest results show that this model can accurately capture the volatility of return rate and produce reliable VaR forecasts.

With the acceleration of globalization, Chinese mainland stock market will indeed be influenced by the stock markets of other countries or regions in the world. Among the selected stock markets, Hong-Kong has maintained a strong and stable positive correlation with Chinese mainland stock market, most likely because the Hong-Kong stock market is still closely linked to mainland due to various geographical and political issues. Japan also has the strongest correlation with Chinese mainland stock market because they both are part of the east Asian economic circle, but the correlation is also the most volatile perhaps for a variety of geographic and political reasons. The United Kingdom and France are part of Europe, they are across the ocean from China, hence the correlations are modest and the volatilities are not particularly dramatic. What is most surprising is that the correlation between the United States and Chinese mainland stock market is minimal and the fluctuations are quite dramatic.

All in all, independent analysis and comprehensive analysis are indispensable in the current era of global integration to conduct economic activities through the Chinese mainland stock market.

5 Acknowledgement

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A Figures and Tables

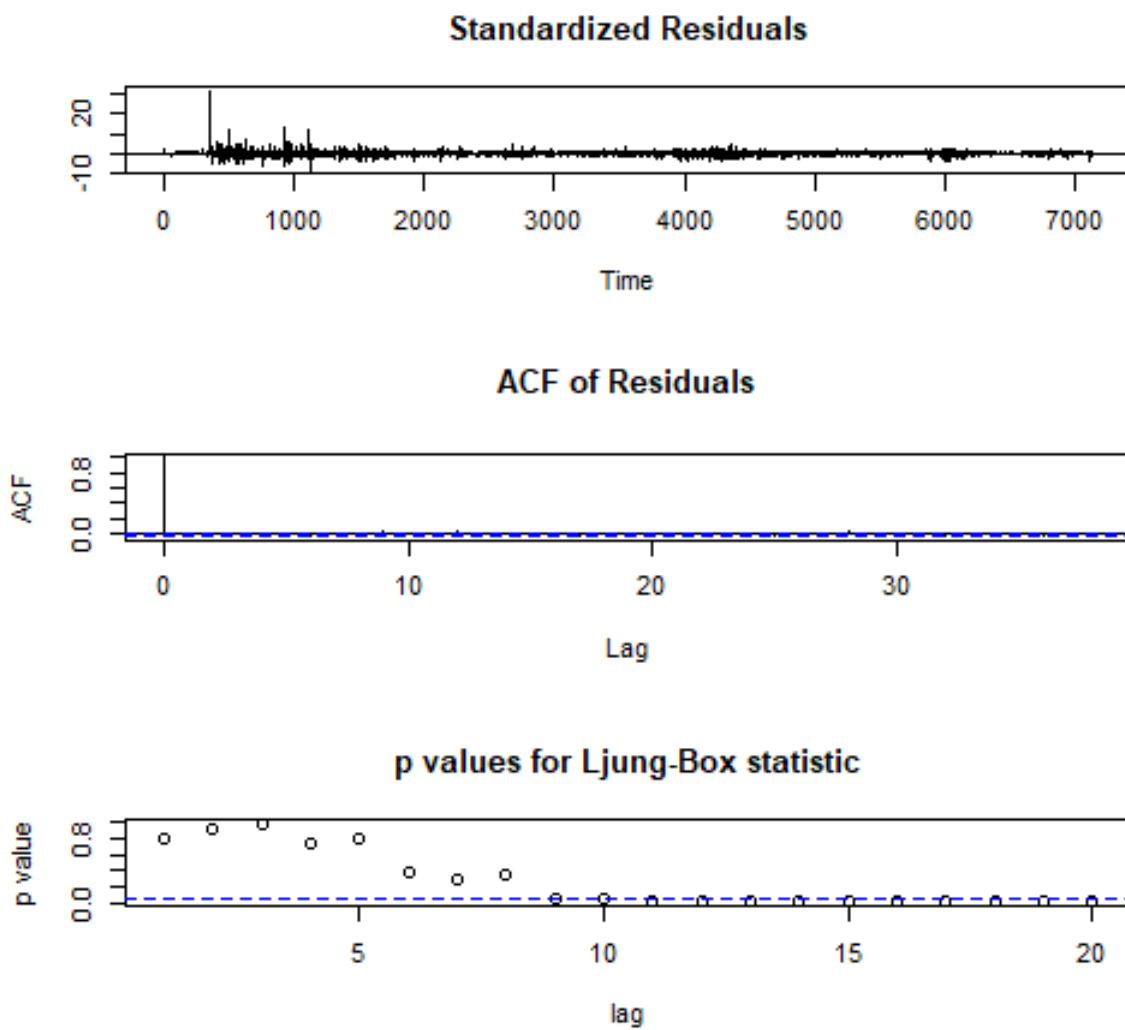


Figure 11: Diagnostic Plot

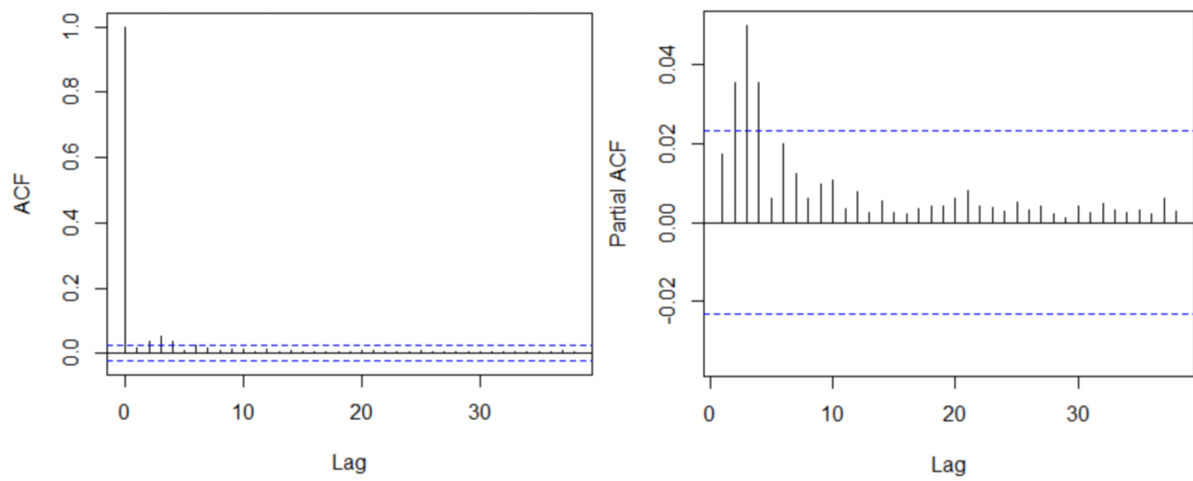


Figure 12: Diagnostic Plot

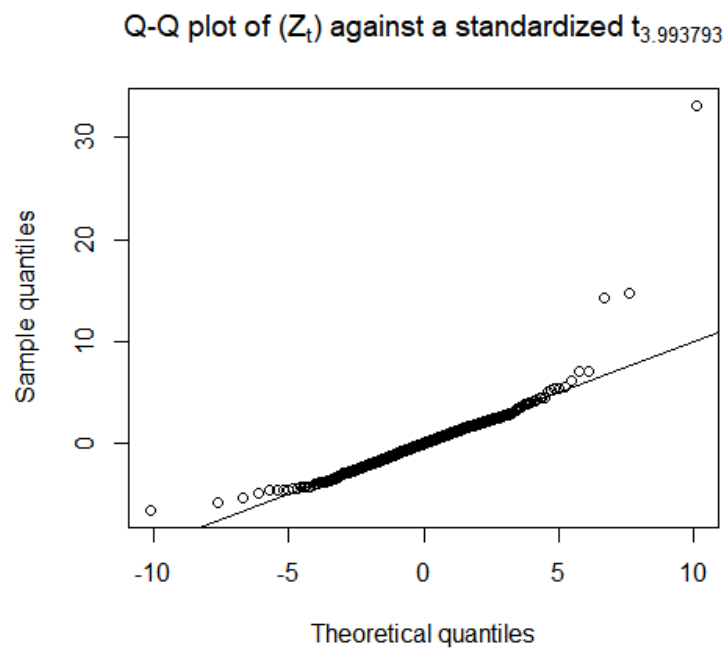


Figure 13: QQ plot of residuals

Index	Order
Chinese mainland	ARMA(3,3)
Hong-Kong	ARMA(4,3)
USA	ARMA(3,3)
Japan	ARMA(3,3)
U.K	ARMA(4,4)
France	ARMA(3,3)

Table 5: ARMA models of six time series

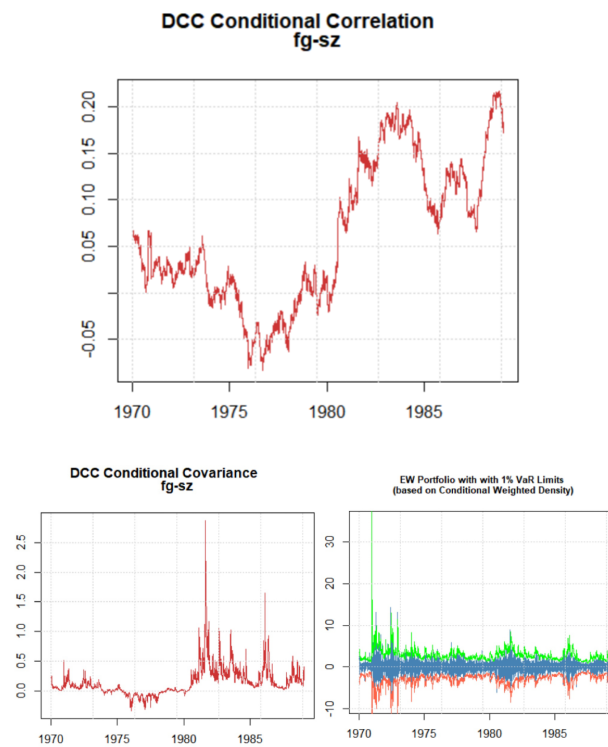


Figure 14: Between Chinese mainland and France

B Code

```
library(tidyverse)
library(lubridate)
library(quantmod)
library(reshape)
library(forecast)
library(stargazer)
library(tseries)
library(moments)
library(fGarch)
library(rugarch)
library(rmgarch)
library(xts)
library(qrmtools)
library(ggplot2)
library(dplyr)
library(backtest)

#### Univariate
### Data
log.return <- function(x){
  c(NA, diff(log(x)))
}
sz <- read.csv("D:\\thesis\\shangzheng.csv",header = FALSE,sep = ",")
sz <- rename(sz,c(V1 = "time"))
sz <- rename(sz,c(V2 = "price"))
sz <- sz %>%
  mutate(sz=round(100*log.return(price), 2))
sz$time <- lubridate::ymd(sz$time)

### Model
```

```

##SZ Return timeseries
Rt1 <- ts(na.omit(sz[, 3]), start = 1, frequency = 1, names = "Rt")
plot(ts(Rt1, freq=1), type='l', xlab='time', ylab='Return')

##ARMA
#test data
adf.test(Rt1)
kpss.test(Rt1)
Box.test(Rt1, lag=12, type="Ljung-Box")
Box.test(Rt1, lag=24, type="Ljung-Box")

#ACF and PACF
tsdisplay(Rt1)

#ARMA and AIC
R1 <- arima(Rt1, order = c(3,0,3))
AIC(R1)
BIC(R1)
summary(R1)

#test ARMA
Box.test(R1$residuals, lag=6, type="Ljung-Box")
Box.test(R1$residuals, lag=8, type="Ljung")
tsdiag(R1, gof=20)

##ARMA-GARCH
##ARCH
#ACF and PACF
acf((R1$residuals)^2, main="")
Pacf((R1$residuals)^2, main="")

#LB test

```

```
Box.test((R1$residuals)^2, lag=20, type="Ljung")
```

```
##GARCH
```

```
skewness(R1$residuals)
```

```
kurtosis(R1$residuals)
```

```
jarque.bera.test(R1$residuals)
```

```
A1 <- ugarchspec(mean.model = list(armaOrder = c(3,3),  
                                include.mean=FALSE),  
                variance.model = list(garchOrder = c(1,1),  
                                model = "sGARCH"), distribution.model = "norm")
```

```
AA1 <- ugarchfit(A1, data=Rt1, fit.control=list(scale=TRUE))
```

```
A2 <- ugarchspec(mean.model = list(armaOrder = c(3,3),  
                                include.mean=FALSE),  
                variance.model = list(garchOrder = c(1,1),  
                                model = "sGARCH"), distribution.model = "std")
```

```
AA2 <- ugarchfit(A2, data=Rt1, fit.control=list(scale=TRUE))
```

```
plot(AA2, which=9)
```

```
A3 <- ugarchspec(mean.model = list(armaOrder = c(3,3),  
                                include.mean=FALSE),  
                variance.model = list(garchOrder = c(1,1),  
                                model = "sGARCH"), distribution.model = "sstd")
```

```
AA3 <- ugarchfit(A3, data=Rt1, fit.control=list(scale=TRUE))
```

```
plot(AA3, which=9)
```

```
##GARCH-M
```

```
AM <- ugarchspec(mean.model = list(armaOrder = c(3,3), archm = TRUE,  
                                archpow = 2),  
                variance.model = list(garchOrder = c(1,1),
```

```

      model = "sGARCH"), distribution.model = "sstd")
AAM <- ugarchfit(AM, data=Rt1, fit.control=list(scale=TRUE))

```

#E-GARCH

```

EA <- ugarchspec(mean.model = list(armaOrder = c(3,3),
      include.mean=FALSE),
      variance.model = list(garchOrder = c(1,1),
      model = "eGARCH"), distribution.model = "sstd")
EAA <- ugarchfit(EA, data=Rt1, fit.control=list(scale=TRUE))

```

```

plot(EAA, which=9)

```

##VaR

```

EA1 <- ugarchspec(mean.model = list(armaOrder = c(3,3),
      include.mean=FALSE),
      variance.model = list(garchOrder = c(1,1),
      model = "eGARCH"), distribution.model = "std")
EAA1 <- ugarchfit(EA1, data=Rt1, fit.control=list(scale=TRUE))
CV <- sigma(EAA1)
M <- fitted(EAA1)
RE <- EAA1@fit$residuals

```

#Plot standardized residuals Z

```

Z <- EAA1@fit$z
stopifnot(all.equal(Z, as.numeric(RE/CV)))
n <- EAA1@fit$coef["shape"]
qq_plot(Z, FUN = function(p) sqrt((n-2)/n) * qt(p, df = n),
      main = substitute("Q-Qplot of ~"*Z[t]*")
      against_a_standardized"~t[n], list(n = n)))

```

```

#Plot VaR and Rt
#0.05
VaR <- as.numeric(M + CV * sqrt((n-2)/n) * qt(0.05, df = n))
VaRR <- as.numeric(M + CV*qdist("std",p=0.05,
                                shape = coef(EAA1)["shape"]))
stopifnot(all.equal(VaR, VaRR))

plot(as.matrix(Rt1), type = "l", xlab = "t",
      ylab = "95%_VaR_value", col = "darkgray")
lines(VaR, col = adjustcolor("red", alpha.f = 0.5))
legend("topright", bty = "n", lty = c(1,1),
      col = c("darkgray", adjustcolor("red", alpha.f = 0.5)),
      legend = c(expression(R[t]), expression(VaR[t])))

#0.01
VaR1 <- as.numeric(M + CV * sqrt((n-2)/n) * qt(0.01, df = n))
VaRR1 <- as.numeric(M + CV*qdist("std",p=0.01,
                                shape = coef(EAA1)["shape"]))
stopifnot(all.equal(VaR1, VaRR1))

plot(as.matrix(Rt1), type = "l", xlab = "t",
      ylab = "99%_VaR_value", col = "darkgray")
lines(VaR1, col = adjustcolor("blue", alpha.f = 0.5))
legend("topright", bty = "n", lty = c(1,1),
      col = c("darkgray", adjustcolor("blue", alpha.f = 0.5)),
      legend = c(expression(R[t]), expression(VaR[t])))

#VaR ugarchroll for forecasting and backtesting
#0.05
EE_ <- ugarchroll(EA1, data = Rt1, n.start = 5000, n.ahead = 1,

```

```

forecast.length = 2134, refit.every = 1,
refit.window = "recursive", solver = "hybrid",
fit.control = list(), solver.control = list(),
calculate.VaR = TRUE, VaR.alpha = 0.05,
keep.coef = TRUE)
REE <- report(EE_, type = "VaR", VaR.alpha = 0.05, conf.level = 0.95)
plot(EE_, which = 4, VaR.alpha = 0.05)
legend("topright", bty="n", legend = "alpha_0.05")

#0.01
EE1 <- ugarchroll(EA1, data = Rt1, n.start = 5000, n.ahead = 1,
forecast.length = 2134, refit.every = 1,
refit.window = "recursive", solver = "hybrid",
fit.control = list(), solver.control = list(),
calculate.VaR = TRUE, VaR.alpha = 0.01,
keep.coef = TRUE)
REE1 <- report(EE1, type = "VaR", VaR.alpha = 0.01, conf.level = 0.99)
plot(EE1, which = 4, VaR.alpha = 0.01)
legend("topright", bty = "n", legend = "alpha_0.01")

```

Multivariate

Data

```

bp <- read.csv("D:\\thesis\\bp.csv", header = FALSE, sep = ",")
bp <- rename(bp, c(V1 = "time"))
bp <- rename(bp, c(V2 = "price"))
bp <- bp %>%
  mutate(bp=round(100*log.return(price), 2))
bp$time <- lubridate::ymd(bp$time)

rj <- read.csv("D:\\thesis\\rj.csv", header = FALSE, sep = ",")
rj <- rename(rj, c(V1 = "time"))

```

```

rj <- rename(rj ,c(V2 = "price"))
rj <- rj %>%
  mutate(rj=round(100*log . return ( price ) , 2))
rj$time <- lubridate::ymd(rj$time)

hs <- read.csv("D:\\thesis\\hs.csv",header = FALSE,sep = ",")
hs <- rename(hs ,c(V1 = "time"))
hs <- rename(hs ,c(V2 = "price"))
hs <- hs %>%
  mutate(hs=round(100*log . return ( price ) , 2))
hs$time <- lubridate::ymd(hs$time)

fg <- read.csv("D:\\thesis\\fg.csv",header = FALSE,sep = ",")
fg <- rename(fg ,c(V1 = "time"))
fg <- rename(fg ,c(V2 = "price"))
fg <- fg %>%
  mutate(fg=round(100*log . return ( price ) , 2))
fg$time <- lubridate::ymd(fg$time)

yg <- read.csv("D:\\thesis\\yg.csv",header = FALSE,sep = ",")
yg <- rename(yg ,c(V1 = "time"))
yg <- rename(yg ,c(V2 = "price"))
yg <- yg %>%
  mutate(yg=round(100*log . return ( price ) , 2))
yg$time <- lubridate::ymd(yg$time)

D1 <-merge(sz ,bp ,by="time" , all=T)
D2 <-merge(fg ,yg ,by="time" , all=T)
D3 <-merge(hs ,rj ,by="time" , all=T)
D4 <-merge(D1 ,D2 ,by="time" , all=T)
DATA <-merge(D3 ,D4 ,by="time" , all=T)

```

```
DATA <-DATA[ ,c(-2,-4,-6,-8,-10,-12)]
```

```
summary(yg)
```

```
skewness(na.omit(yg$yg))
```

```
kurtosis(na.omit(yg$yg))
```

```
###Model
```

```
##Return timeseries
```

```
HongKong <- ts(na.omit(DATA[,2]), start = 1, frequency = 1 )
```

```
plot(ts(HongKong, freq=1), type='l', xlab='time', ylab='HongKong')
```

```
R2 <- arima(HongKong, order = c(4,0,3))
```

```
Japan <- ts(na.omit(DATA[, 3]), start = 1, frequency = 1 )
```

```
plot(ts(Japan, freq=1), type='l', xlab='time', ylab='Japan')
```

```
R3 <- arima(Japan, order=c(3,0,3))
```

```
Chinese <- ts(na.omit(DATA[, 4]), start = 1, frequency = 1 )
```

```
plot(ts(Chinese, freq=1), type='l', xlab='time', ylab='Chinese')
```

```
USA <- ts(na.omit(DATA[, 5]), start = 1, frequency = 1 )
```

```
plot(ts(USA, freq=1), type='l', xlab='time', ylab='USA')
```

```
R4 <- arima(USA, order = c(3,0,3))
```

```
France <- ts(na.omit(DATA[, 6]), start = 1, frequency = 1 )
```

```
plot(ts(France, freq=1), type='l', xlab='time', ylab='France')
```

```
R5 <- arima(France, order = c(3,0,3))
```

```
UK <- ts(na.omit(DATA[, 7]), start = 1, frequency = 1 )
```

```
plot(ts(UK, freq=1), type='l', xlab='time', ylab='UK')
```

```
R6 <- arima(UK, order = c(4,0,4))
```


##DCC-GARCH

```
garch1 <- ugarchspec(mean.model = list(armaOrder = c(3,3),  
                                     include.mean = FALSE),  
                    variance.model = list(garchOrder = c(1,1),  
                                         model = "eGARCH"), distribution.model = "std")
```

```
garch2 <- ugarchspec(mean.model = list(armaOrder = c(4,3),  
                                     include.mean = FALSE),  
                    variance.model = list(garchOrder = c(1,1),  
                                         model = "eGARCH"), distribution.model = "std")
```

```
garch3 <- ugarchspec(mean.model = list(armaOrder = c(4,4),  
                                     include.mean = FALSE),  
                    variance.model = list(garchOrder = c(1,1),  
                                         model = "eGARCH"), distribution.model = "std")
```

```
DGS1 <- dccspec(uspec = multispec(replicate(2,garch1)),  
              dccOrder = c(1,1), distribution = "mvt")
```

```
DGS2 <- dccspec(uspec = multispec(c(garch1,garch2)),  
              dccOrder = c(1,1), distribution = "mvt")
```

```
DGS3 <- dccspec(uspec = multispec(c(garch1,garch3)),  
              dccOrder = c(1,1), distribution = "mvt")
```

#HongKong

```
HK <- dccfit(DGS2, na.omit(DATA[,c(2,4)]),  
            fit.control = list(eval.se=TRUE))
```

```
plot(HK, which=3)
```

```
plot(HK, which=4)
```

```
plot(HK, which=5)
```

```
HKF <- dccforecast(HK, n.ahead = 100)
```

```
plot(HKF, which=3)
```

```
#USA
```

```
US <- dccfit(DGS1, na.omit(DATA[,c(4,5)]),  
             fit.control = list(eval.se=TRUE))
```

```
plot(US, which=3)
```

```
plot(US, which=4)
```

```
plot(US, which=5)
```

```
USF <- dccforecast(US, n.ahead = 100)
```

```
plot(USF, which=3)
```

```
#Japan
```

```
JA <- dccfit(DGS1, na.omit(DATA[,c(3,4)]),  
             fit.control = list(eval.se=TRUE))
```

```
plot(JA, which=3)
```

```
plot(JA, which=4)
```

```
plot(JA, which=5)
```

```
JAF <- dccforecast(JA, n.ahead = 100)
```

```
plot(JAF, which=3)
```

```
#UK
```

```
UK <- dccfit(DGS3, na.omit(DATA[,c(4,7)]),  
             fit.control = list(eval.se=TRUE))
```

```
plot(UK, which=3)
```

```
plot(UK, which=4)
```

```
plot(UK, which=5)
```

```
UKF <- dccforecast(UK, n.ahead = 100)
```

```

plot(UKF, which=3)

#France
FA <- dccfit(DGS1, na.omit(DATA[,c(4,6)]),
             fit.control = list(eval.se=TRUE))
plot(FA, which=3)
plot(FA, which=4)
plot(FA, which=5)

FAF <- dccforecast(FA, n.ahead = 100)
plot(FAF, which=3)

```